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To Combat Extreme Heat, Let's Invest in Better Walking and Cycling Networks

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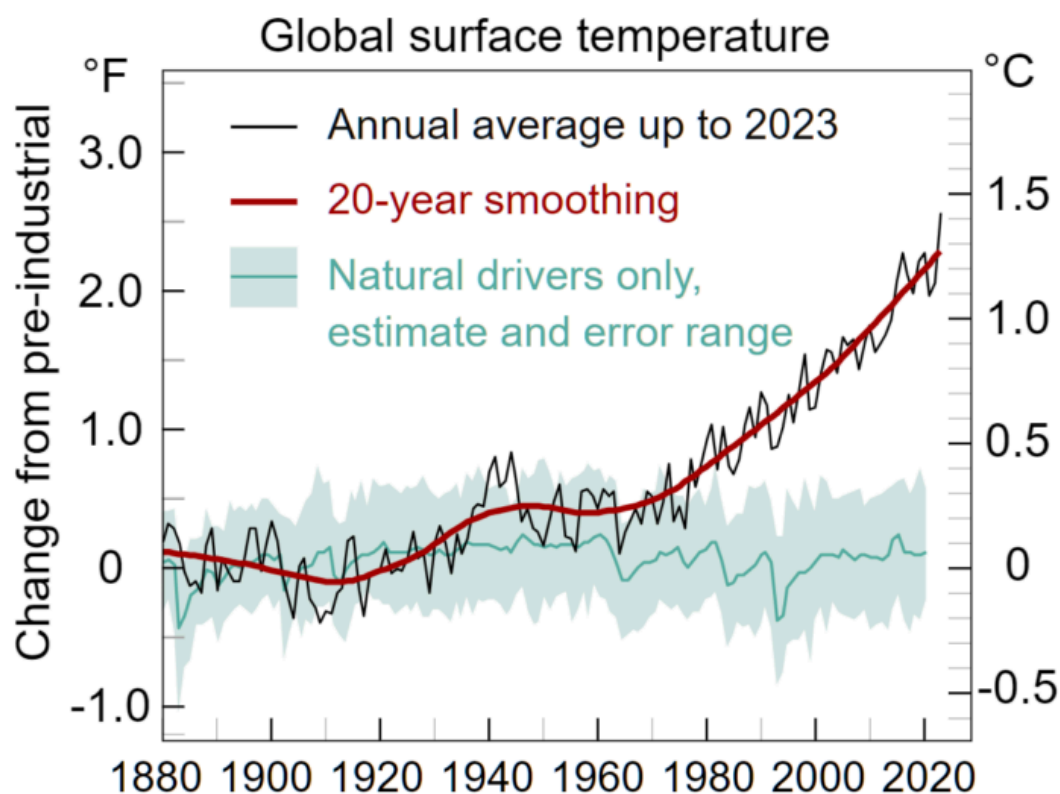
Between March and June of this year, extreme heat took its toll across India. Temperatures climbed to 50 °C (122 °F) in Eastern India during the region's longest-ever recorded heat wave. In the city of New Delhi, almost 200 unhoused people died from heat-related illness in one week in June according to Reuters. Outdoor workers in the city suffered too: "My body can't take it, but I have to keep cycling," said one bicycle rickshaw driver in a 2024 CNN report. As witnessed in New Delhi, moving around outdoors becomes increasingly unsafe as today's cities heat up from climate change. But that does not have to be the

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case for the cities of tomorrow. It is both possible and crucial to invest in heat-safe walking and cycling infrastructure to save lives. What's more, broader walking and cycling investments can help mitigate the deadly cycle of climate change and heat waves.

Cities around the world are now hotter than ever. In addition to global warming, urban areas tend to be warmer than nearby suburban and rural ones because they have less green space and more heat-absorbing buildings and roads. Extreme urban temperatures can be **especially dangerous** for people walking and cycling, including those who use wheelchairs or other mobility aids. Heat stress combined with physical exertion can lead to dehydration, heatstroke, and cardiovascular strain.

Certain populations are more severely exposed to heat and are more vulnerable to that exposure: that includes the young and the elderly, lower-income people, marginalized racial communities, people with disabilities, unhoused people, and outdoor workers. Many people in these groups rely on walking and cycling outdoors as an affordable and accessible mode of transport. Additionally, marginalized and lower-income neighborhoods **tend to have less green space**, more concrete and asphalt, and fewer health and infrastructural resources. All of this means more exposure to unsafe outdoor conditions in the heat.



Global surface temperatures on the planet have risen sharply in recent decades as greenhouse gas emissions skyrocket. Graph: NASA.gov

To combat the dangers of walking and cycling while hot, especially for heat-vulnerable communities, cities need to employ both immediate and sustainable long-term solutions in the built environment. There are plenty of short-term strategies to upgrade existing infrastructure to be more heat resilient. A range of heat-focused, cost-effective street improvements are emerging around the world, including:

- Preserving and planting native trees and vegetation that cool and shade people walking and cycling, especially in areas that have less green space
- Coating roads and pathways with lighter-colored coatings that absorb less heat, and using heat-resilient construction materials
- Installing inexpensive shade tarps
- Using permeable pavers that allow water to filter through, reducing ground heat
- Sheltering outdoor public transport stops
- Adding misting systems, which cool down the microclimate of walking and cycling paths
- Providing free hydration supplies and heat safety education during particularly hot periods
- Ensuring public access to potable drinking water

It is worth noting that cities have existed in **hot climates for centuries**, primarily with narrow, shaded, and nature-integrated streets that stay cool naturally. Modern cities, however, have given way to **vehicle-focused streets and infrastructure** that have made heat-intensive tar, asphalt, and concrete much more dominant. As we plan for a hotter future, resilient street improvements can include the tried-and-true techniques of ancient cities with newer technologies. They should also always prioritize the populations and areas that are the most vulnerable to extreme heat.



This year, extreme heatwaves swept major cities like New Delhi, India, and compounded mobility challenges. Image: ITDP Global

To achieve this, cities need to broadly increase walking and cycling investments in order to make active mobility easier, safer, cooler, and more accessible in the long term. **Compact city planning** centered around active mobility options plays a large part in lowering urban temperatures. This is because sprawling, car-centered development eats up green space and replaces it with large swaths of heat-absorbing roads and parking lots. **A 2010 study** of 83 cities in the United States even found that the most sprawling cities had the greatest increases in extreme heat events over time.

Additionally, pedestrians and cyclists in sprawling cities must travel further to reach healthcare, jobs, and essential destinations. In compact cities centered on green infrastructure, people can spend less time traveling in unsafe heat conditions and still access the places and services that are necessary during heat waves. When it is safer and easier to walk and cycle, more people will get out of their cars and fewer will buy cars in the first place, helping address a main contributor to climate change. **ITDP's own research** has shown that up to 29% of pedestrians and up to 16% of cyclists on new, high-quality paths have shifted away from using high-polluting private vehicles. With the rise of e-bikes, cycling can certainly become an attractive and less strenuous travel option, even on hotter days.

This modal shift results in cleaner air, safer streets, accessible transport networks, individual cost savings, and fewer emissions. By building and improving over 100 kilometers of footpaths, for example, **Chennai, India** prevented between 4,200 and 12,000 tonnes of emissions per year. The extensive protected cycling networks in **Guangzhou, China** and **Bogotá, Colombia** save about 16,000 and 22,000 tonnes per year, respectively. If more cities replicate this work, they can also begin to reduce the vast amounts of planet-warming gases produced by road transport.



Wide, shaded paths in Yichang, China, are used by pedestrians and cyclists of all ages. Image: ITDP China

Some cities are already moving towards both short- and long-term improvements. Since 2013, the city of Yichang, China has built **a vast network** of bicycle lanes, footpaths, and green spaces in collaboration with ITDP China, including over 192 kilometers of linear parks. This infrastructure includes heat safety features, such as shaded pedestrian islands near BRT stations. Along one avenue, the city built covered walkways and planted more than 700 trees.

The city’s investments are helping create a safer environment for pedestrians and cyclists while encouraging more people to choose active mobility.

With global temperatures still poised to spike, our cities cannot wait any longer to address the impacts of extreme heat on pedestrians and cyclists. For people who lack other transport options, public resources like increased shade, seating, drinking water, and beyond can quite literally be lifesaving. More broadly, building bike- and pedestrian-friendly cities is critical to lowering emissions and helping governments reach their climate goals.



The dire heatwaves of the last few years should be wake-up call for every city to invest in better walking and cycling infrastructure today.



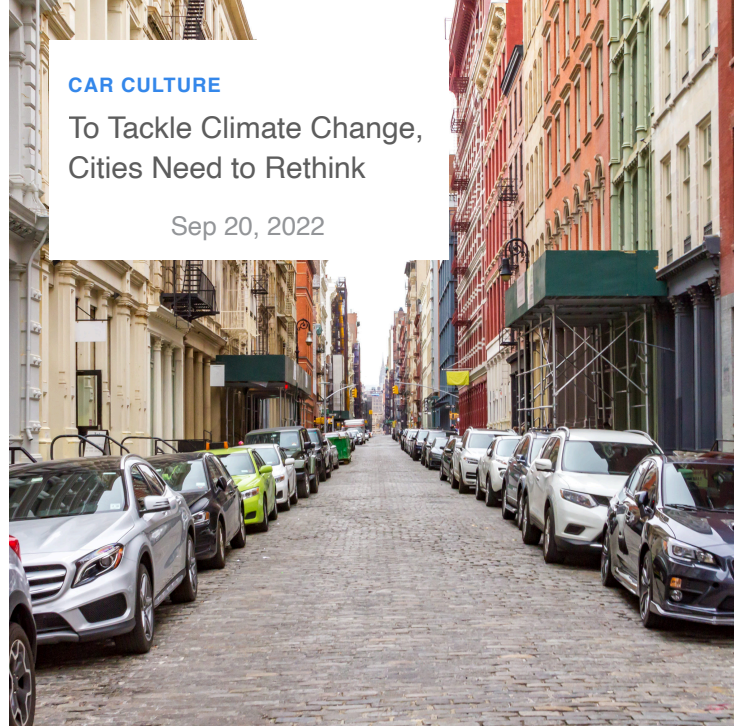
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